Contents

Código de las figuras del libro

Marcos Bujosa Brun

July 10, 2026

1 Figuras con fuente tikz

1.1 Correlación

```
\documentclass[border=10pt]{standalone}
\usepackage[utf8]{inputenc}
\usepackage{nacal}
\usepackage{pgfplots}
\usepackage{tikz-3dplot}

\pgfplotsset{compat=newest}
\usepackage{tikz}
\usetikzlibrary{matrix,decorations.pathreplacing}
% Añadimos arrows.meta para dar soporte a los estilos de vectores con flechas stealth
\usetikzlibrary{calc,angles,positioning,intersections,quotes,decorations.markings,arrows.meta}
\usepackage{tkz-euclide}

\pgfplotsset{width=10cm,height=6cm}

% 1. DEFINIMOS LA VISTA 3D (Ángulos de rotación para los ejes)
\tdplotsetmaincoords{55}{65}

\begin{document}
  «Colores»

  \begin{tikzpicture} [scale=3,
    tdplot_main_coords,
    axis/.style={-,black, thin},
    vector/.style={-stealth},
    vector guide/.style={dashed,gray,very thin}]

    % standard tikz coordinate definition using x, y, z coords
    \tdplotsetcoord{P}{1.2}{50}{70}
    \tdplotsetcoord{Q}{1}{-70}{-140}
    \tdplotsetcoord{C}{1.9}{45}{40}
    \tdplotsetcoord{O}{0}{0}{0}

    \tdplotsetrotatedcoords{0}{0}{-45}

    \tdplotsetcoord{X}{.7}{90}{0}
    \tdplotsetcoord{XX}{.8}{-90}{0}
    \tdplotsetcoord{Y}{1.2}{90}{90}    % vectores constantes
    \tdplotsetcoord{YY}{.3}{90}{-90}
    \tdplotsetcoord{Z}{1.0}{0}{0}
    \tdplotsetcoord{ZZ}{-.45}{0}{0}     % vertical
    \tdplotsetcoord{H}{1}{-90}{-60}

    % draw axes
    \draw[axis]  (XX) -- (X);
    \draw[axis,color=blue!95!black,thin]  (YY) -- (Y) node[color=blue!90!black, anchor=west,font=\footnotesize]{$\Span{\Vect{1}}
    \draw[axis]  (ZZ) -- (Z);

    \coordinate (Ymedia) at (0,0.865,0);
    \coordinate (Yhat)   at (Pxy);

    \coordinate (Xmedia) at (0,0.605,0);

    \tkzMarkRightAngle[size=.08,gray](O,Pxz,P)
    \tkzMarkRightAngle[size=.08,gray](O,Qxz,Q)

    \draw[vector guide,gray!50!black]         (Pxz) -- (P);
    \draw[vector guide]         (Ymedia) -- (P);

    \draw[vector guide,gray!50!black]         (Qxz) -- (Q);
    \draw[vector guide]         (Xmedia) -- (Q);

    % draw a vector from O to P
```

```
    \draw[vector, red!30!gray] (O) -- (P) node[anchor=south west]{$\Vect{v}$};
    \draw[vector, color=blue!20!gray]  (O) -- (Ymedia) node[anchor=south west]{}; %${\Vect{\Media{v}}}$};
    \draw[vector, red!50!black,thick]  (O) --  (Pxz)node[anchor=south, xshift=0.55mm]{$\Vect{v}-\Vect{\Media{v}}$};

    \draw[vector, green!30!gray] (O) -- (Q) node[anchor=west]{$\Vect{x}$};
    \draw[vector, color=blue!30!gray]  (O) -- (Xmedia) node[anchor=south]{}; %${\Vect{\Media{x}}}$};
    \draw[vector, green!40!black,thick]  (O) --  (Qxz)node[anchor=north west]{$\Vect{X}-\Vect{\Media{x}}$};

    \draw pic[draw=black,angle radius=17,angle eccentricity=1.4,gray!50!black,thick] {angle = Qxz--O--Pxz} node[anchor=south wes

  \end{tikzpicture}
\end{document}
```

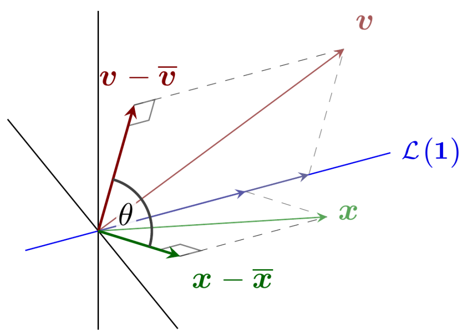

Figure 1: Interpretación geométrica de la correlación como coseno de un ángulo

1.2 Correlación 2

```
\documentclass[border=10pt]{standalone}
\usepackage[utf8]{inputenc}
\usepackage{nacal}
\usepackage{pgfplots}
\usepackage{tikz-3dplot}

\pgfplotsset{compat=newest}
\usepackage{tikz}
\usetikzlibrary{matrix,decorations.pathreplacing}
% Añadimos arrows.meta para dar soporte a los estilos de vectores con flechas stealth
\usetikzlibrary{calc,angles,positioning,intersections,quotes,decorations.markings,arrows.meta}
\usepackage{tkz-euclide}

\pgfplotsset{width=10cm,height=6cm}

% 1. DEFINIMOS LA VISTA 3D (Ángulos de rotación para los ejes)
\tdplotsetmaincoords{90}{0}

\begin{document}
  «Colores»

  \begin{tikzpicture} [scale=3,
    tdplot_main_coords,
    axis/.style={-,black, thin},
    vector/.style={-stealth},
    vector guide/.style={dashed,gray,very thin}]

    % standard tikz coordinate definition using x, y, z coords
    \tdplotsetcoord{P}{1.2}{50}{70}
```

```
    \tdplotsetcoord{Q}{1}{-70}{-140}
    \tdplotsetcoord{C}{1.9}{45}{40}
    \tdplotsetcoord{O}{0}{0}{0}

    \tdplotsetrotatedcoords{0}{0}{-45}

    \tdplotsetcoord{X}{1.0}{90}{0}
    \tdplotsetcoord{XX}{.1}{-90}{0}
    \tdplotsetcoord{Y}{1.2}{90}{90}    % vectores constantes
    \tdplotsetcoord{YY}{.3}{90}{-90}
    \tdplotsetcoord{Z}{0.9}{0}{0}
    \tdplotsetcoord{ZZ}{-.1}{0}{0}     % vertical
    \tdplotsetcoord{H}{1}{-90}{-60}

    % draw axes
    \draw[axis]  (XX) -- (X);
    \draw[axis]  (ZZ) -- (Z);

    \coordinate (Ymedia) at (0,0.865,0);
    \coordinate (Yhat)   at (Pxy);

    \coordinate (Xmedia) at (0,0.605,0);

    \draw[vector guide,gray!50!black]         (Pxz) -- (P);
    \draw[vector guide]         (Ymedia) -- (P);

    \draw[vector guide,gray!50!black]         (Qxz) -- (Q);
    \draw[vector guide]         (Xmedia) -- (Q);

    \draw[vector, red!50!black,thick]  (O) --  (Pxz)node[anchor=south, xshift=0.55mm]{$\Vect{v}-\Vect{\Media{v}}$};

    \draw[vector, green!40!black,thick]  (O) --  (Qxz)node[anchor=north west]{$\Vect{X}-\Vect{\Media{x}}$};

    \draw pic[draw=black,angle radius=17,angle eccentricity=1.4,gray!50!black,thick] {angle = Qxz--O--Pxz} node[anchor=south wes

  \end{tikzpicture}
\end{document}
```

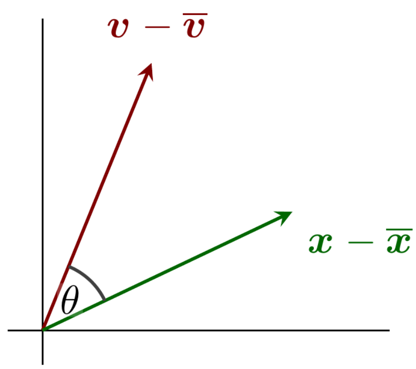

Figure 2: Interpretación geométrica de la correlación como coseno de un ángulo

1.3 Varianza

```
\documentclass[border=10pt]{standalone}
\usepackage[utf8]{inputenc}
```

```
\usepackage{nacal}
\usepackage{pgfplots}
\usepackage{tikz-3dplot}

\usepackage{tikz}
\usetikzlibrary{matrix,decorations.pathreplacing}
% Añadida explícitamente arrows.meta para asegurar que Stealth funcione siempre
\usetikzlibrary{calc,angles,positioning,intersections,quotes,decorations.markings,arrows.meta}
\usepackage{tkz-euclide}

\pgfplotsset{width=10cm,height=6cm}

\begin{document}
  «Colores»

  \begin{tikzpicture}

    % Definir los vectores
    \coordinate (O) at (0,0);            % Origen
    \coordinate (U) at (1,1);            % Vector Uno
    \coordinate (X) at (1,5);            % Vector de Datos
    % \coordinate (X) at (2,5);            % Vector de Datos

    % El código ($(O)!(X)!(U)$) significa: el punto de la recta que une (O) y (U) que está más cerca de (X)

    \coordinate (M) at ($(O)!(X)!(U)$);  % Vector mu * Uno.

    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    %%%% Elementos por detrás de la figura principal %%%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

    % Dibujar cuadrícula
    \draw[step=1cm,lightgray,very thin] (-.4,-.4) grid (6.3,5.8);
    % Dibujar ejes
    \draw[-] (-.6,0) -- (6.3,0);   % eje x
    \draw[-] (0,-.6) -- (  0,5.3); % eje y

    % Etiquetas de los vectores x y mu*Uno
    \node[blue, right, fill=white,opacity=.9,text opacity=1] at (X) [above]{\large$\boldsymbol{y}$};
    \node[green!50!black,rotate=45,below left, fill=white,opacity=.9,text opacity=1] at (M) {\large$\mu_{\boldsymbol{y}}\cdot\bo
    \node[purple,rotate=45,below,fill=white,opacity=.9,text opacity=1]  at ($(O)!0.5!(U)$) {\footnotesize$\boldsymbol{1}=(1,1)$}

    \node[gray]  at (-1.6,-1.5) {\large Estadística versus geometría};
    \node[gray]  at (-1.6,-2) {(o la varianza vs el T$^{ma}$ de Pitágoras)};


    % Dibujar la recta generada por el vector (1,1)
    \draw[-,thick,gray] (-.3,-.3) -- (4.3,4.3) node[right]{};

    % Marca de ángulo recto
    \tkzMarkRightAngle[fill=blue!10,size=.4](O,M,X)

    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    %%%% Cuadrados de las longitudes %%%%%%%%%%%%%%%%%%%
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

    % Varianza
    \tkzDefSquare(X,M);
    \path (X) -- (tkzFirstPointResult) node[midway,fill=white,opacity=.7,text opacity=1] {\Large$\sigma_{\boldsymbol{y}}^2$};
    \tkzDrawPolygon[thin, orange, fill=orange,opacity=.1](X, M, tkzFirstPointResult, tkzSecondPointResult)
    % \path (tkzSecondPointResult) -- (tkzFirstPointResult) node[midway,sloped,purple,above] {\slshape $\boldsymbol{\lambda}$aura

    % Cuadrado de la longitud de x
    \tkzDefSquare(O,X); \tkzDrawPolygon[thin, blue, fill=blue,opacity=.1](O, X, tkzFirstPointResult, tkzSecondPointResult)
    \path (O) -- (tkzFirstPointResult) node[midway,text opacity=1] {\Large$\|\boldsymbol{y}\|_s^2$};
    % \path (tkzSecondPointResult) -- (O)  node[midway,sloped,green!50!black,below] {\slshape $\boldsymbol{\beta}$ujos$\alpha$};

    % Cuadrado de la longitud de mu*Uno
```

```
    \tkzDefSquare(M,O);
    \path (M) -- (tkzFirstPointResult) node[midway,fill=white,opacity=.95,text opacity=1] {\Large$\|\boldsymbol{\mathop{\overlin
    \tkzDrawPolygon[thin, green, fill=green,opacity=.1](M, O, tkzFirstPointResult, tkzSecondPointResult)
    % \path (tkzSecondPointResult) -- (tkzFirstPointResult) node[midway,sloped,blue,below] {\slshape $\delta$el Río};

    % DIBUJAR LOS VECTORES
    % Dibujar el vector X
    \draw[->,very thick, blue] (O) -- (X); % node[above,fill=white,opacity=.9,text opacity=1]{$\boldsymbol{y}$};

    % Dibujar el vector mu * Uno
    \draw[->,very thick,green!50!black] (O) -- (M); % node[right, fill=white,opacity=.9,text opacity=1]{$\mu\boldsymbol{1}$};

    % Dibujar la línea punteada entre X y mu*Uno
    \draw[dashed] (X) -- (M);

    % Dibujar el vector Uno
    \draw[->,very thick, purple] (O) -- (U); % node[sloped,midway,below,fill=white,opacity=.5,text opacity=1]{$\boldsymbol{1}=(1,

  \end{tikzpicture}
\end{document}
```

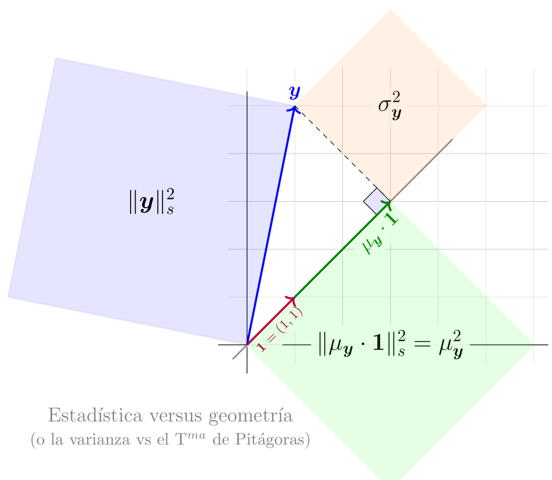

Figure 3: Interpretación geométrica de la varianza mediante el teorema de Pitágoras

2 Trozos de código

Preámbulo para generar un fichero con únicamente una figura, y que use la notación de `nacal`

```
\documentclass[border=10pt]{standalone}
\usepackage[utf8]{inputenc}
\usepackage{nacal}
```

Contenido del preámbulo para generar figuras en 3d con tikz (con proyecciones ortogonales usando la librería tikz-3dplot.

```
\usepackage{pgfplots}
\usepackage{tikz-3dplot}

%\pgfplotsset{compat=newest}
```

```
\usepackage{tikz}
\usetikzlibrary{matrix,decorations.pathreplacing}
\usetikzlibrary{calc,angles,positioning,intersections,quotes,decorations.markings}
\usepackage{tkz-euclide}
```

Contenido del preámbulo para normalizar la proporción alto ancho de la figura

```
%\pgfplotsset{width=10cm,height=6cm,compat=1.8}
\pgfplotsset{width=10cm,height=6cm}
```

Código para la normalización de colores en las figuras

```
\def\Rojo{red!50!black}
\def\RojoOscuro{red!20!black}
\def\RojoClaro{red!90!black}
\def\Verde{green!50!black}
\def\VerdeOscuro{green!30!black}
\def\VerdeClaro{green!50!gray}
\def\Azul{blue!50!black}
\def\AzulOscuro{blue!35!black}
\def\AzulClaro{blue!60!gray}
```

3 Compilación

Para obtener los ficheros en emacs hay que teclear `C-c C-v t`

La conversión de pdf a png se realiza con `poppler-utils` (especificamente con `pdftoppm`). Así que hay que instalar el programa con `aptitude` (o programa similar).

```
%.pdf : %.tex
	pdflatex $<
	rm $(basename $<).aux
	rm $(basename $<).log

%.png : %.pdf
	pdftoppm -r 500 $< $@ -png
	mv $@-1.png ../$(basename $@).png
	#mv $@-1.png ../png/$(basename $@).png

tex_FILES := $(wildcard *.tex)
png_FILES := $(tex_FILES:.tex=.png)

pngs: $(png_FILES)

pdfs:
	latexmk -pdf
	latexmk -c
```

Para generar los pngs teclear dentro del directorio `tex`

```
make pngs
```

Si además se quiere generar los pdf en la carpeta `tex` (aunque no es necesario) se puede teclear `make pdfs`

```
make pdfs
```

4 Ajustes de las figuras

```
convert Correlacion.png Correlacion2.png -gravity center +append -background white -gravity center -extent 2500x${ALTURA} Correl
```

```
convert Varianza.png -gravity center -background white -extent 4500x Varianza_ancho.png
```